Sam Gunn

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Positions

Massachusetts Institute of Technology
C.L.E. Moore Instructor of Mathematics *Summer 2026–*

Education

The University of California, Berkeley Advisor: Umesh Vazirani
Ph.D. in Computer Science *Fall 2020–Summer 2026*

The University of Texas at Austin Advisors: Scott Aaronson and Cameron Gordon
B.S. in Mathematics *Fall 2016–Spring 2020*

Selected works

1. **How to sketch a learning algorithm**. *In submission*. arXiv.
2. **Improved pseudorandom codes from permuted puzzles**. With Miranda Christ, Noah Golowich, Ankur Moitra, and Daniel Wichs. *In submission*. ePrint.
3. **Black-box crypto is useless for pseudorandom codes**. With Sanjam Garg and Mingyuan Wang. *TCC 2025 (outstanding paper award)*. ePrint.
4. **An undetectable watermark for generative image models**. With Xuandong Zhao and Dawn Song. *ICLR 2025*. arXiv.
5. **SoK: Watermarking for AI-generated content**. With Xuandong Zhao et al. *S&P 2025*. arXiv.
6. **Ideal pseudorandom codes**. With Omar Alrabiah, Prabhanjan Ananth, Miranda Christ, and Yevgeniy Dodis. *STOC 2025 (invited to the SICOMP Special Issue)*. ePrint.
7. **Quantum one-time protection of any randomized algorithm**. With Ramis Movassagh. *CRYPTO 2025*. ePrint.
8. **Classical commitments to quantum states**. With Yael Kalai, Anand Natarajan, and Agi Villanyi. *QIP 2025, STOC 2025*. ePrint.
9. **Pseudorandom error-correcting codes**. With Miranda Christ. *CRYPTO 2024*. ePrint.
10. **Undetectable watermarks for language models**. With Miranda Christ and Or Zamir. *COLT 2024, RWC 2024*. ePrint.
11. **How to use quantum indistinguishability obfuscation**. With Andrea Coladangelo. *STOC 2024*. ePrint.
12. **Approaching the quantum Singleton bound with approximate error correction**. With Thiago Bergamaschi and Louis Golowich. *QIP 2024, STOC 2024*. arXiv.
13. **Commitments to quantum states**. With Nathan Ju, Fermi Ma, and Mark Zhandry. *QIP 2023, STOC 2023 (invited to the SICOMP Special Issue)*. ePrint.

Recognition

1. Google PhD Fellowship 2022–2025
2. Lotfi A. Zadeh Prize 2024–2025
3. Simons Institute annual magazine: link
4. Communications of the ACM: link
5. TCC 2025 Outstanding Paper Award
6. Invitations to the SICOMP Special Issues for STOC 2023 and 2025 (reserved for the top 10% of accepted STOC papers)

Talks

1. **How to sketch a learning algorithm**
   - MIT Crypto+AI seminar (May 2026).
   - Institute for Advanced Study (May 2026).
   - Flatiron Institute (June 2026).
   - STOC Theory for Trustworthy and Interpretable AI Workshop (June 2026).
   - Invited talk at ITC (August 2026).
2. **Undetectable backdoors** at the UT Austin Theory and Alignment group (November 2025).
3. **Pseudorandom error-correcting codes**
   - NTT Research (March 2024).
   - Google Research NYC (April 2024).
   - CRYPTO 2024 (August 2024).
   - NYU (March 2025).
   - Harvard Theory Seminar (September 2025).
   - TCC 2025 (December 2025).
4. **Undetectable watermarks for language models**
   - Google Research Mountain View (July 2023).
   - MIT Cryptography Group (August 2023).
   - NYU Cryptography Seminar (September 2023). Video here.
   - Berkeley Machine Learning Theory Seminar (September 2023).
5. **Quantum one-time protection of any randomized algorithm** at CRYPTO 2025 (August 2025).
6. **Quantum indistinguishability obfuscation**
   - CIQC Spark Talk, March 2024.
   - The University of Ottawa, April 2024.
7. **Approaching the quantum Singleton bound with approximate error correction** at the Simons Quantum Colloquium (December 2022). Video here.
8. **Commitments to quantum states** at STOC 2023 (June 2023). Video here.
9. **Unclonable cryptography** guest lecture for John Wright's graduate quantum complexity course at UC Berkeley (December 2022).
10. **Certified entropy from random circuit sampling** at Berkeley Theory Lunch (March 2022).
11. **On the classical hardness of spoofing linear cross-entropy benchmarking** plenary talk at SQuInT 2020 (February 2020).